

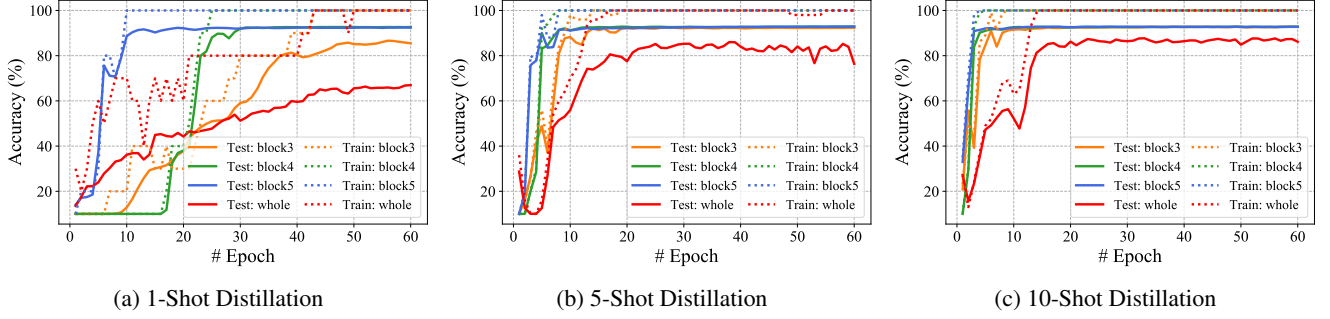


Figure 3: The comparison of optimization efficiency on block grafting and the whole student on CIFAR10.

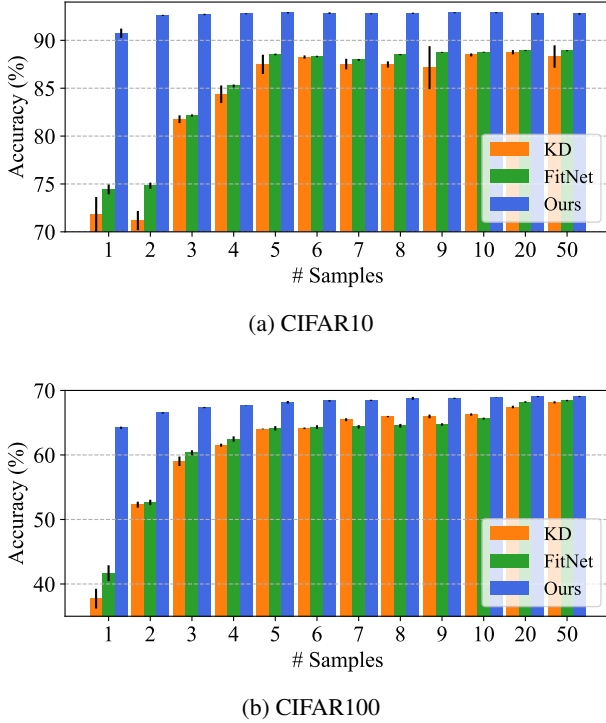


Figure 4: Learning with different numbers of samples. All experiments are averaged over five runs and error bars correspond to the standard deviation.

beled samples per class. This is achieved by a dual-stage approach. In the first stage, the student is split into blocks and grafted onto the corresponding position of teacher. The student blocks are optimized with fixed teacher blocks using distillation loss. In the second stage, the trained student blocks are incrementally grafted onto the teacher and trained to connect to each other, until the whole student network replaces the teacher one. Experimental results on several benchmarks demonstrate that the proposed method successfully transfers knowledge from teacher in few-shot setting and achieves comparable performance as knowledge distillation using full dataset.

	original	block1	block2	block3	block4
Params (M)	/	0.22→0.14	1.12→0.52	6.82→2.10	13.11→8.40
	/	36.4%↓	53.6%↓	69.2%↓	35.9%↓
Accuracy (%)	73.31	72.55	70.33	68.11	68.40

Table 3: The performance of block grafting from ResNet18 to ResNet34 on ILSVRC-2012. “original” denotes the original teacher network.

Few-shot distillation can significantly reduce the training cost of neural network. For future work, we plan to explore network architecture search on distillation using block grafting, which aims to find a series of more efficient block modules. We believe that this work is one step toward efficient network architecture search.

Acknowledgements. This work is supported by National Natural Science Foundation of China (U20B2066, 61976186), the Major Scientific Research Project of Zhejiang Lab (No. 2019KD0AC01) and Alibaba-Zhejiang University Joint Research Institute of Frontier Technologies.

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